

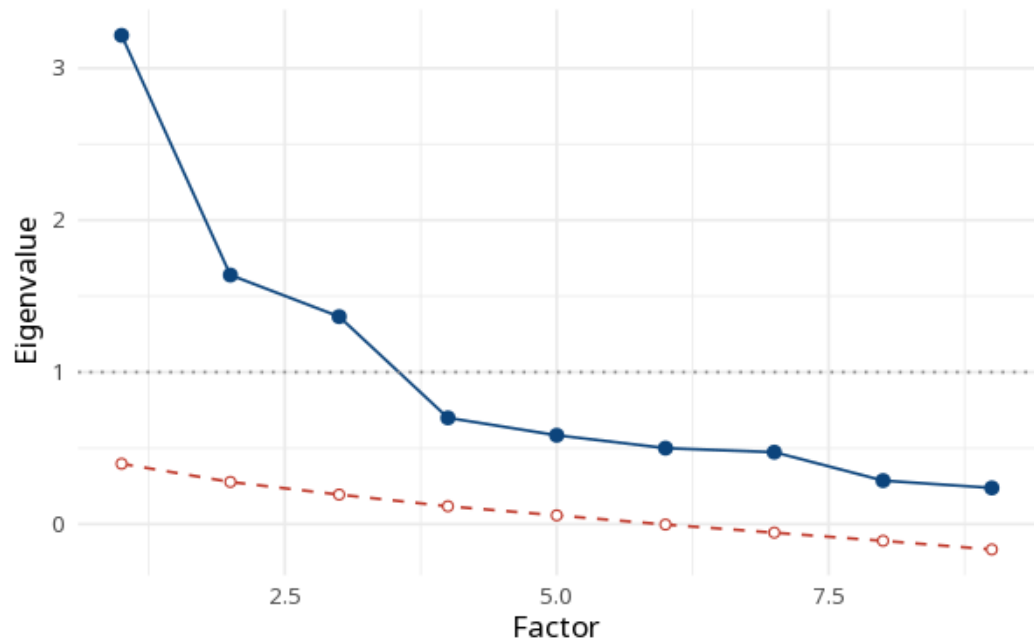
1 Exploratory factor analysis (EFA)

KMO	Bartlett's χ^2	df	p
.75	904.1	36	<.001

Factor	Eigenvalue	% variance	Cumulative %
F1	2.24	24.9	24.9
F2	1.34	14.9	39.8
F3	1.28	14.2	54.0

Item	F1	F2	F3	Communality
x1		.59		.48
x2		.51		.26
x3		.69		.45
x4	.85			.73
x5	.89			.75
x6	.81			.69
x7			.73	.51
x8			.69	.52
x9		.38	.46	.46

	F1	F2	F3
F1	1.00		
F2	0.32	1.00	
F3	0.21	0.26	1.00



Solid: observed · Dashed: parallel analysis 95th pct · Dotted: eigenvalue = 1

Check KMO ($> .60$ acceptable, $> .70$ preferred; Kaiser & Rice, 1974) and a significant Bartlett's test ($p < .001$) for suitability. Parallel analysis (Horn, 1965) is the recommended retention rule — it corrects the upward bias of the Kaiser eigenvalue > 1 rule (Zwick & Velicer, 1986). Table 3 shows the rotated pattern matrix: each loading is the unique contribution of the factor to that item. Loadings $|\leq .32|$ ($\sim 10\%$ variance) are suppressed; a loading $\geq .32$ on two or more factors flags a cross-loader (Tabachnick & Fidell). For oblique rotation (oblimin), the matrix shows interfactor correlations — if all $|\leq .32|$, orthogonal rotation (varimax) may be preferred. EFA is exploratory — it discovers structure; to confirm a proposed factor structure run a CFA, ideally on a separate sample (Watkins, 2018).

EFA (KMO = .75, Bartlett's $\chi^2(36) = 904.1$, $p < .001$) with parallel analysis retained 3 factors explaining 54.0% of variance (oblimin rotation).

Statistical basis: KMO sampling-adequacy verbal labels ($\geq .60$ acceptable, $\geq .70$ preferred) (Kaiser, H. F., Rice, J. (1974). "Little Jiffy, Mark IV." Educational and Psychological Measurement, 34(1), 111-117.); Parallel analysis as the preferred factor-retention rule (Horn, J. L. (1965). "A rationale and test for the number of factors in factor analysis." Psychometrika, 30(2), 179-185.); Kaiser eigenvalue > 1 rule systematically over-extracts (Zwick, W. R., Velicer, W. F. (1986). "Comparison of five rules for determining the number of components to retain." Psychological Bulletin, 99(3), 432-442.); EFA best-practice reporting standard (Watkins, M. W. (2018). "Exploratory Factor Analysis: A Guide to Best Practice." Journal of Black Psychology, 44(3), 219-246.).